



Screen-Based vs. Non-Screen Leisure: How Social Context Shapes Momentary Happiness and Stress

Farangiz Akhadova '27, Data Science Major Capstone

Introduction

- Leisure activities occupy a substantial portion of daily life
- Screen-based leisure (television, computer use, games) may differ from non-screen leisure (reading, hobbies, relaxing) in its effects on well-being
- Social context, being alone versus with others, could change how an activity affects happiness and stress
- The ATUS Well-Being Module (2010, 2012, 2013) randomly selected three activities per respondent and measured momentary happiness and stress on a 0-6 scale.
- Prior work has found associations between screen time and lower psychological well-being (Twenge et al., 2018). Folk, D., & Dunn, E. (2026) found that social interaction is associated with higher happiness across daily activities

Research Question

How do momentary happiness and stress differ between screen-based and non-screen leisure activities, and do those patterns change depending on whether people are alone or with others?

Data

- Used the American Time Use Survey Well-Being Module from 2010, 2012, and 2013.
- The final dataset was merged at the episode level using the activity, well-being, respondent, roster, and Who files.
- The final analytic sample includes 12,444 respondents and 14,986 leisure episodes.

Variables / Sample Description

Key Variables

- Outcomes:** Momentary happiness (WUHAPPY, 0-6), momentary stress (WUSTRESS, 0-6)
- Activity Type:**
 - Screen-based leisure: TV watching (120303), leisure computer use (120308), video games (120307)
 - Non-screen leisure: Reading (120312), arts & crafts (120309), relaxing (120301)
- Social context:** Alone vs. with others (derived from ATUS Who file; code 18 = alone)
- Controls:** Age (TEAGE), sex (TESEX), employment status (TEFES)
- Excluded:** Episodes with invalid well-being values (coded -2 or -3); non-leisure episodes

Sample Description

- Screen-based episodes outnumber non-screen episodes: 11,273 vs. 3,713
- Largest group: Screen-based with others (n = 8,138); Smallest: Non-screen alone (n = 1,262)
- Mean age: 50.3 years (range 15-85); roughly equal sex split
- 81% of respondents contribute 1 episode; max 3 episodes per person (32.7% of episodes come from repeated respondents)

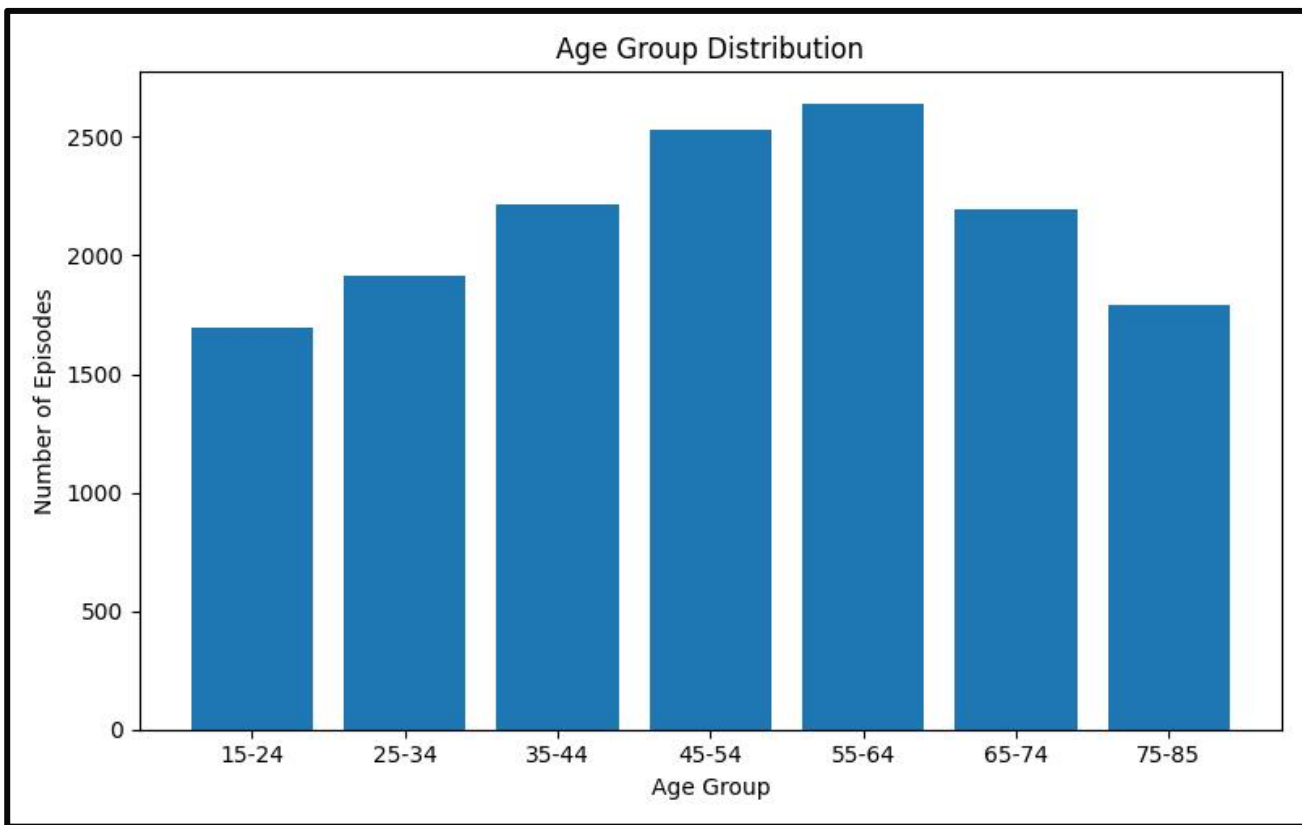


Figure 1. The analytic sample included episodes across all age groups, with the most episodes from ages 45-64.

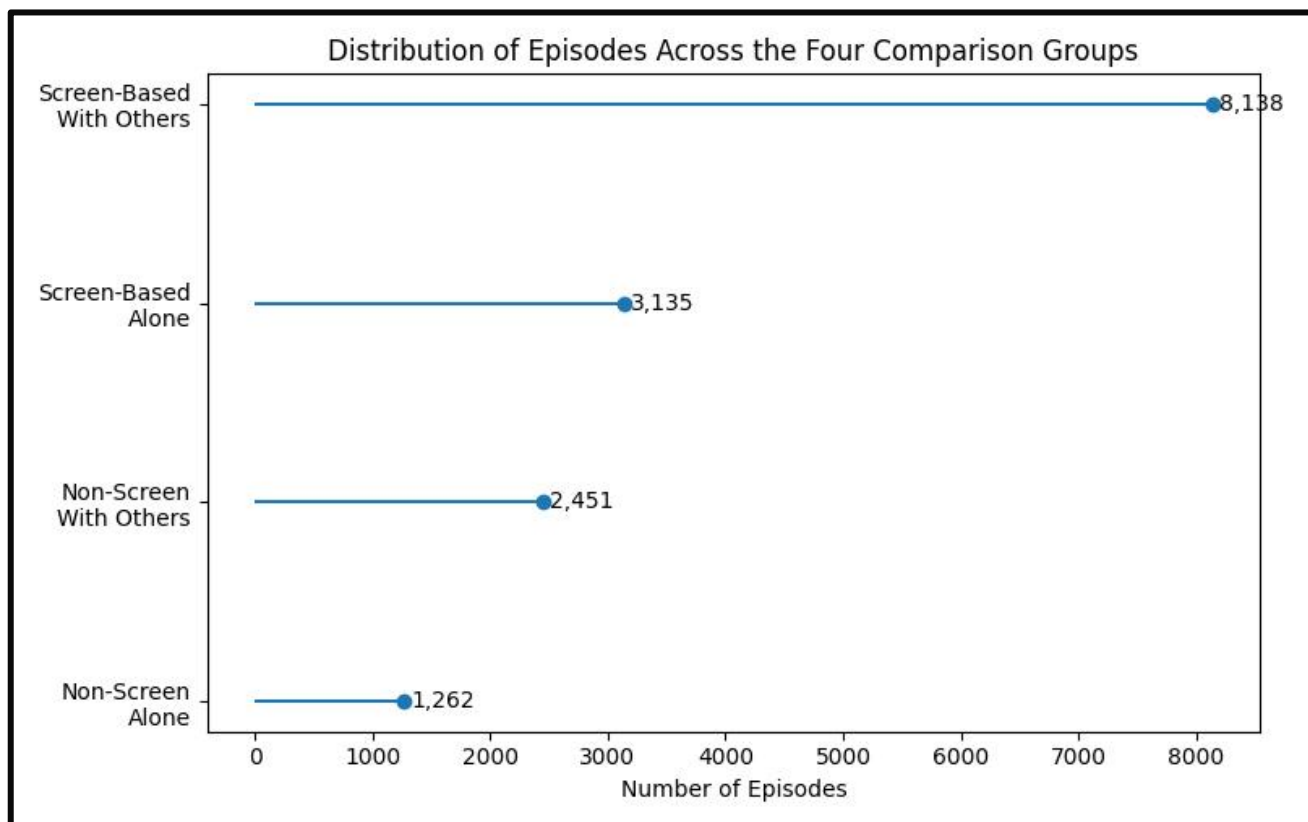


Figure 2. Screen-based leisure with others was the largest group in the sample, while non-screen leisure alone was the smallest.

Methods

Data Pipeline:

- Merged 5 ATUS files; constructed 6-digit activity codes; derived alone/with-others variable

Model 1: Weighted Least Squares (WLS) + Clustered Standard Errors (SEs)

- Applied ATUS activity weights (WUFNACTWTP) for population-level estimates
- Clustered standard errors by respondent ID (TUCASEID) to account for repeated episodes

Model 2: Linear Mixed-Effects Model

- Random intercept per respondent; accounts for within-person correlation structurally
- Tradeoff: mixed model was unweighted due to software limitations.

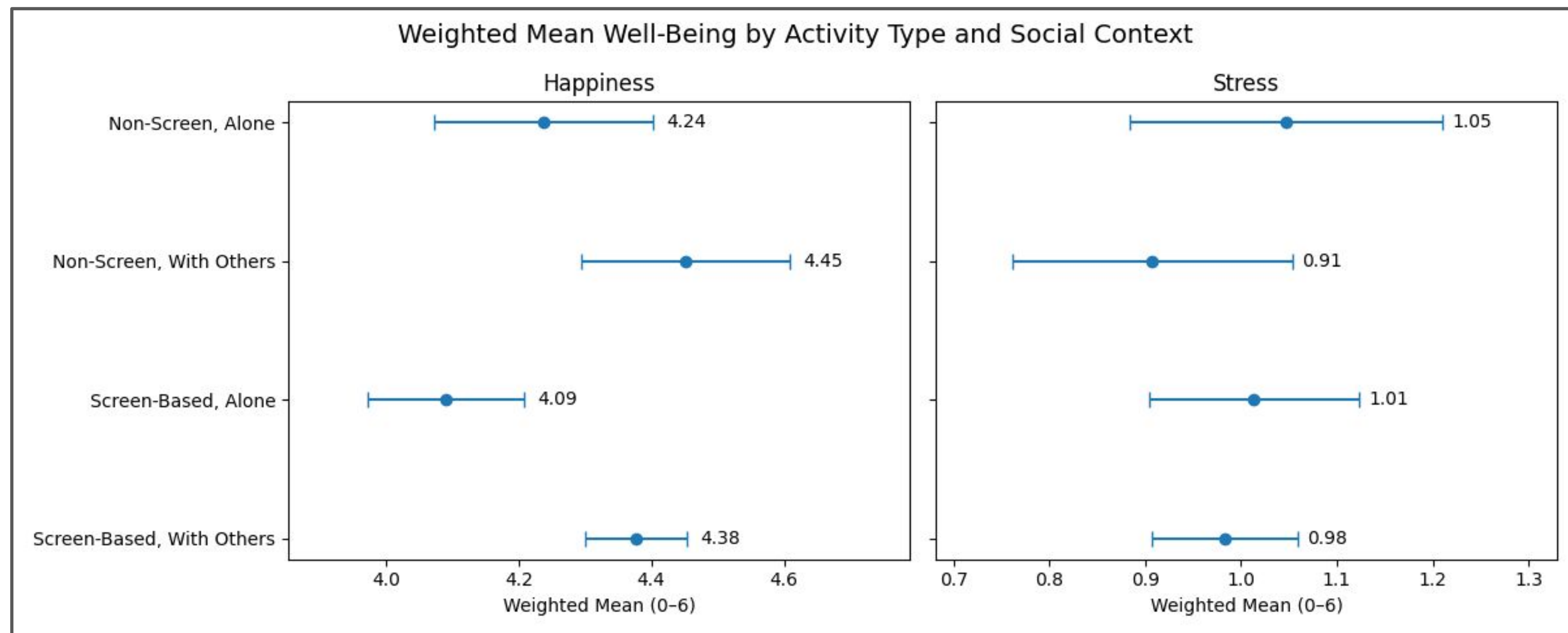


Figure 3. Group differences were clearer for happiness than for stress. Non-screen leisure with others had the highest happiness, while screen-based leisure alone had the lowest.

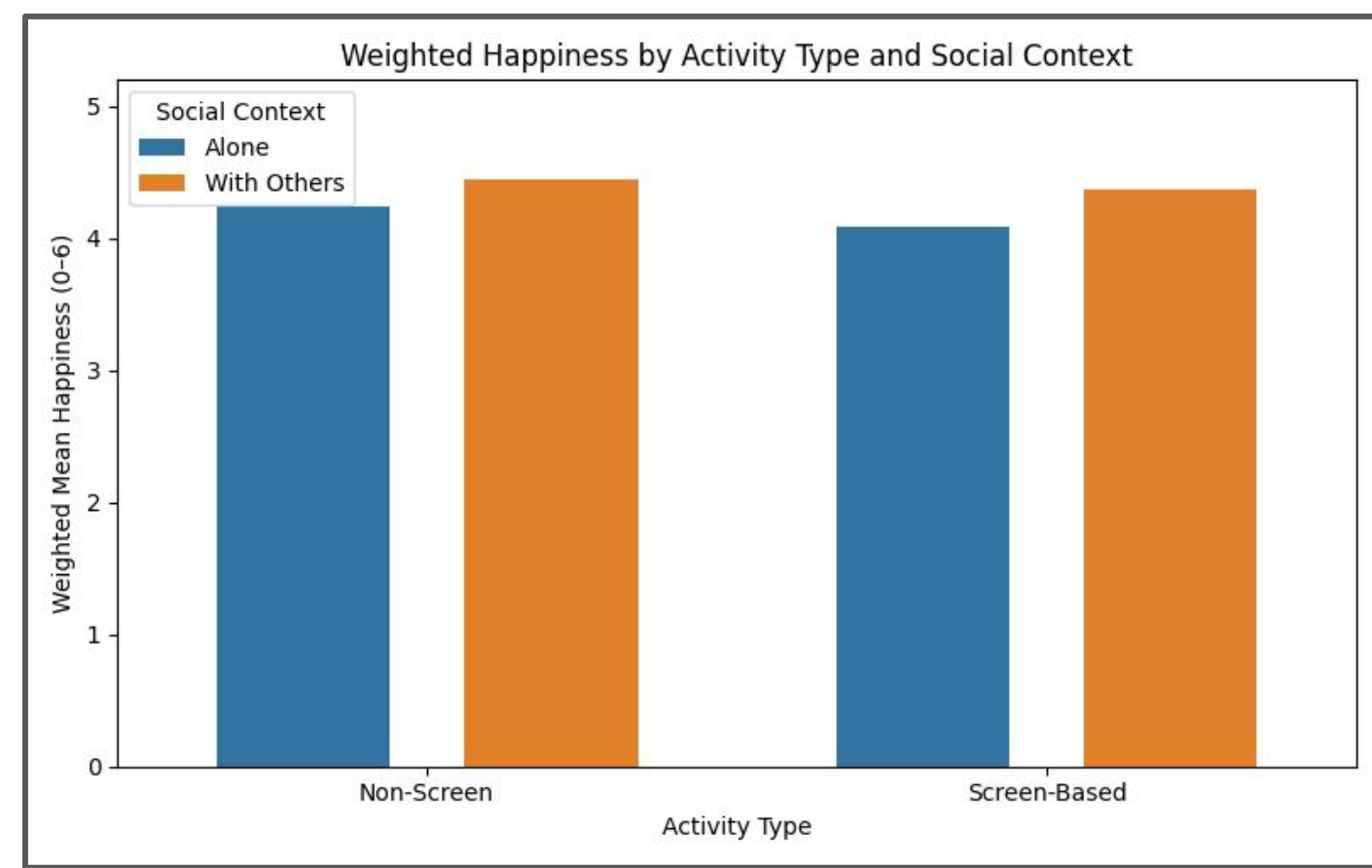


Figure 4. Weighted happiness was highest for non-screen leisure with others and lowest for screen-based leisure alone.

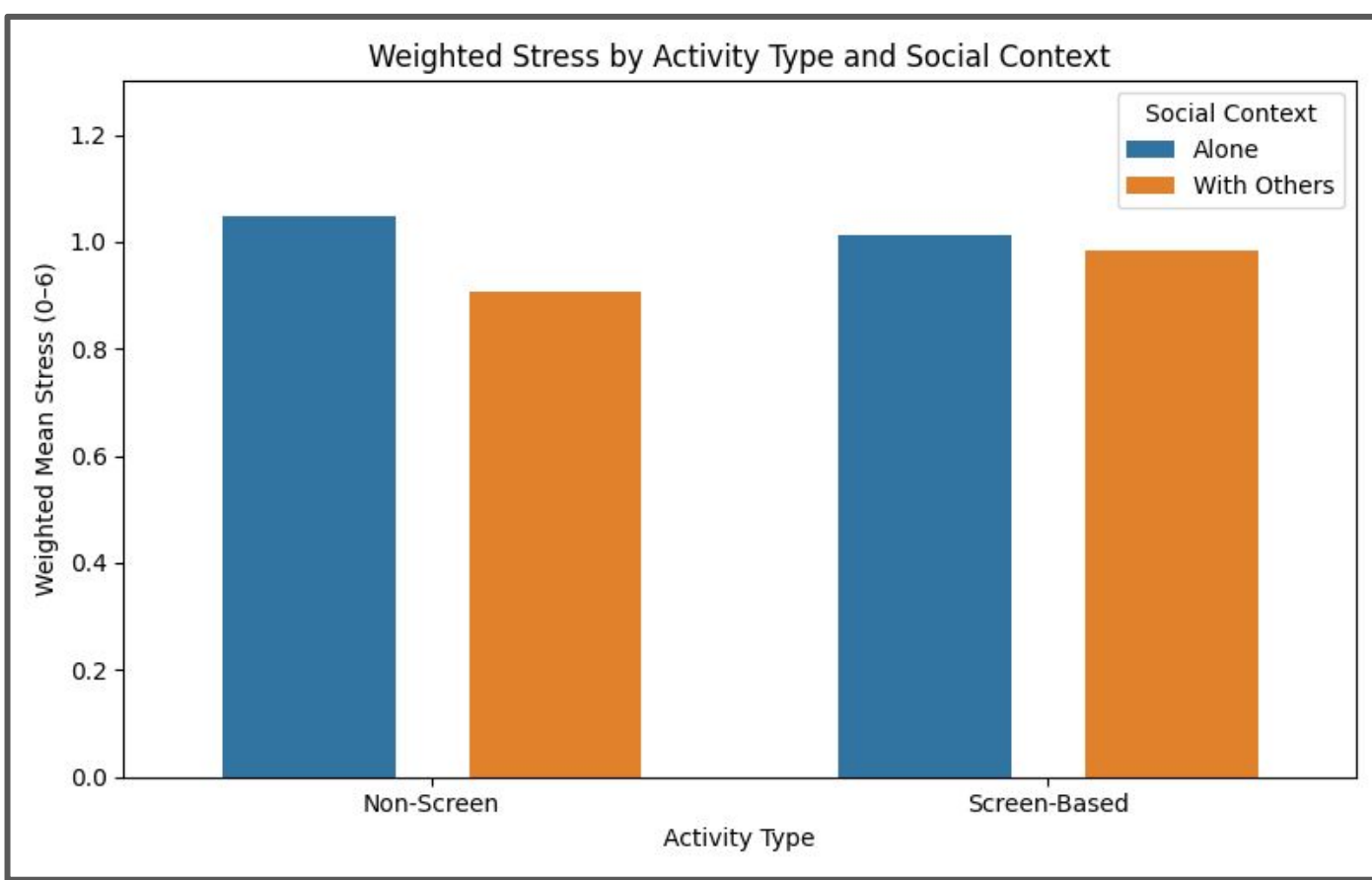


Figure 5. Stress differed less across groups than happiness. Non-screen leisure with others had the lowest weighted stress, while non-screen leisure alone had the highest.

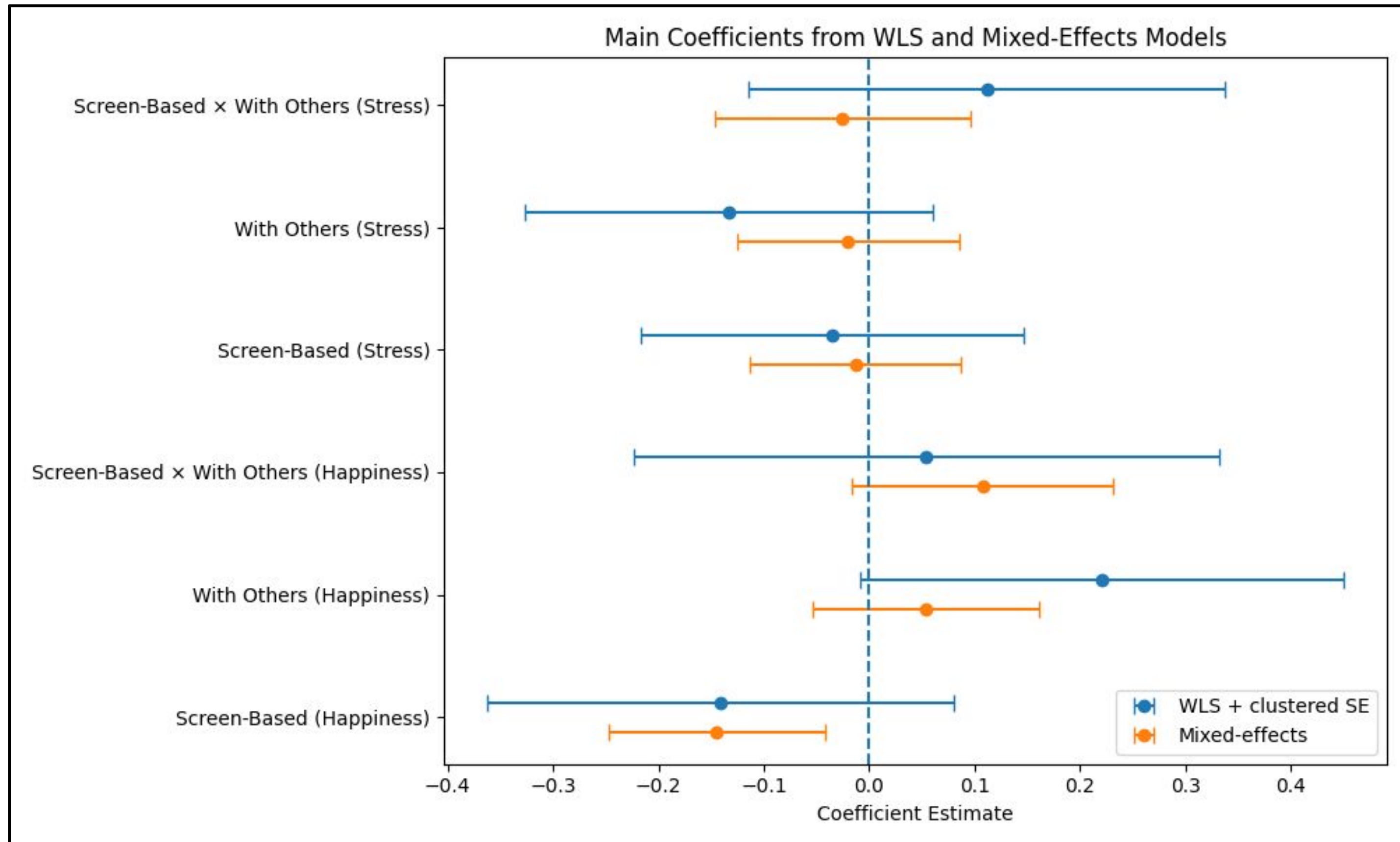


Figure 6. Coefficients were generally clearer for happiness than for stress, and the WLS and mixed-effects models showed similar overall patterns.

Outcome	Term	WLS + clustered SE	Mixed-effects
Happiness	Screen-Based	-0.141 (p=.212)	-0.144** (p=.006)
	With Others	+0.221 (p = .060)	+0.054 (p = .327)
	Screen x Others	+0.054 (p=.701)	+0.108 (p=.087)
	Fit	R ² =.010; Adj. R ² =.009	Marg. R ² =.004; Cond. R ² =.575
Stress	Screen-Based	-0.035 (p=.708)	-0.013 (p=.800)
	With Others	-0.133 (p=.177)	-0.020 (p=.708)
	Screen x Others	+0.112 (p=.332)	-0.025 (p=.682)
	Fit	R ² =.008; Adj. R ² =.007	Marg. R ² =.004; Cond. R ² =.636

Table 1. Model results. Coefficients are relative to the reference group: non-screen leisure, alone, male, employed full-time. Intercepts not shown: WLS = 4.28 happiness, 0.94 stress; mixed-effects = 4.32 happiness, 1.06 stress.

Results and Discussion

Happiness Findings:

- Mixed-Effects Model: Screen-based leisure is associated with a 0.144-point lower happiness score compared to non-screen leisure (95% CI: -0.247 to -0.042, p = 0.006) - a 2.4% reduction on the 0-6 scale. The interaction between activity type and social context is not statistically significant ($\beta = 0.108$, p = 0.087, 95% CI: -0.016 to 0.231). This means the happiness disadvantage of screen-based leisure does not significantly differ depending on whether a person is alone or with others.
- WLS Model: Screen-based leisure shows a consistent direction but larger standard errors: $\beta = -0.141$ (95% CI: -0.363 to 0.080, p = 0.212). Being with others is marginally significant ($\beta = 0.221$, p = 0.060). The interaction is not significant ($\beta = 0.054$, p = 0.701).
- Consistency across models: The screen-based coefficient is nearly identical (-0.144 vs. -0.141)

Stress Findings:

- No predictors showed statistically significant effects on momentary stress. All coefficients have p-values > 0.05 and 95% confidence intervals crossing zero.

Significant Control Variables

- Across both models, being out of the labor force (TEFES=5) was associated with lower happiness (WLS: -0.208, mixed: -0.122) and higher stress (WLS: +0.230, mixed: +0.168).
- Unemployment (TEFES=4) was associated with higher stress in both models (WLS: +0.347, mixed: +0.308) but lower happiness only in the mixed model (-0.136).
- In the mixed model only, females reported higher happiness (+0.099) and higher stress (+0.080), and older age was associated with very slightly lower stress (-0.002).

Model Fit

- WLS R² and adjusted R² are nearly identical for happiness (0.010 vs. 0.009) and stress (0.008 vs. 0.007), with a difference of only 0.001, indicating no overfitting
- Mixed-effects 5-fold cross-validation R² values are near zero (0.003 for happiness, 0.002 for stress) and stable across folds, confirming no overfitting
- Typical prediction error is about 1.66 points on the 0-6 scale (happiness RMSE = 1.657 $\pm$ 0.027, stress RMSE = 1.663 $\pm$ 0.029), with small standard deviations indicating stable performance across data subsets. Marginal R² = 0.004 for both outcomes, meaning activity type and social context alone explain less than 1% of episode-level variance
- Conditional R² = 0.575 for happiness and 0.636 for stress, meaning once stable person-level differences are added, the model explains 58-64% of total variance
- The large gap between marginal and conditional R² shows that who you are matters substantially more than what specific leisure activity you are doing for determining momentary happiness and stress

- VIF < 3 .1.

Model	Training R ²	CV R ² (mean)	CV RMSE (mean $\pm$ SD)
Mixed Happiness	0.004 (marg.)	0.003	1.657 $\pm$ 0.027
Mixed Stress	0.004 (marg.)	0.002	1.663 $\pm$ 0.029
WLS Happiness	.010	-0.001	1.660 $\pm$ 0.035
WLS Stress	0.008	-0.001	1.666 $\pm$ 0.017

Table 2. Model fit and 5-fold cross-validation results. Training R² for mixed-effects models refers to marginal R², meaning fixed effects only. CV R² and RMSE are based on held-out respondents. RMSE is reported in the original 0-6 well-being scale.

Conclusion

- Screen-based leisure is associated with slightly lower momentary happiness; results are clearest in the mixed-effects model
- Social context does not significantly moderate this relationship
- Stress is largely unrelated to leisure type or social context in this data
- Low marginal R² (0.004) with high conditional R² (0.575-0.636) confirms that stable person-level differences drive momentary well-being, not activity type or social context

Limitations

- The analysis is observational; no causal claims
- Well-being ratings capture sampled episodes from one diary day, not long-term leisure patterns
- Social context collapsed to alone vs. with others; relationship quality not captured
- Activity groupings involve judgment calls; different groupings may yield different results
- A methodological tradeoff exists between the two modeling approaches. WLS incorporates survey weights for population representativeness. Mixed-effects models account for within-person dependence (33% of episodes from repeat respondents) but do not use weights due to software limitations.

References

Folk, D., & Dunn, E. (2026). Everything is Better Together: Analyzing the Relationship Between Socializing and Happiness in the American Time Use Survey. *Social Psychological and Personality Science*, 17(4), 426-438.
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